Group Assignment - R and Python

Using the IBM Telco Customer Churn dataset introduced during the lecture, conduct an independent exploratory **analysis in R** (you can find the dataset in the course blackboard or here). The objective is to go beyond the examples discussed in class and investigate the dataset in greater depth. In particular, focus on features and relationships that were not examined, or were only briefly touched upon, during the lecture.

You can find an documentation of the dataset here.

Exploratory Analysis
Explore the dataset independently and identify variables, customer characteristics, and relationships that you consider relevant. Define your own analytical questions and use the dataset to investigate them. Do not simply reproduce the analyses demonstrated during the lecture.

Statistical Analysis
Use appropriate statistical calculations to summarize, compare, and better understand the data. Select the methods according to the questions you choose to investigate, and briefly interpret what the results indicate. The aim is to use statistics to uncover meaningful patterns rather than to demonstrate as many functions as possible.

Data Visualization
Use visualizations to support and communicate your analysis. You may reuse techniques introduced during the lecture, but you should also explore plot types or visualization techniques that were not demonstrated in class. Choose the visualization according to the information you want to communicate and briefly explain the patterns or relationships it reveals. You are encouraged to consult the R and ggplot2 documentation and experiment independently with suitable approaches.

Findings
Conclude the analysis by summarizing the most relevant patterns or insights identified by the group. Discuss what the results may imply for understanding customer churn and indicate any additional questions that emerged from your exploration.

Deliverable
Submit one **R Markdown (.Rmd) document** per group containing executable R code, statistical calculations, visualizations, short explanations and interpretations, and a concise summary of the main findings. The submission should read as a coherent exploratory analysis rather than as a collection of unrelated exercises.

Use of AI Tools
The use of AI tools such as ChatGPT, GitHub Copilot, or similar assistants is allowed. AI may be used to explore R functions, understand documentation, suggest analytical approaches or visualizations, and debug code. However, the group is responsible for critically reviewing generated code and for understanding and being able to explain the submitted analysis.